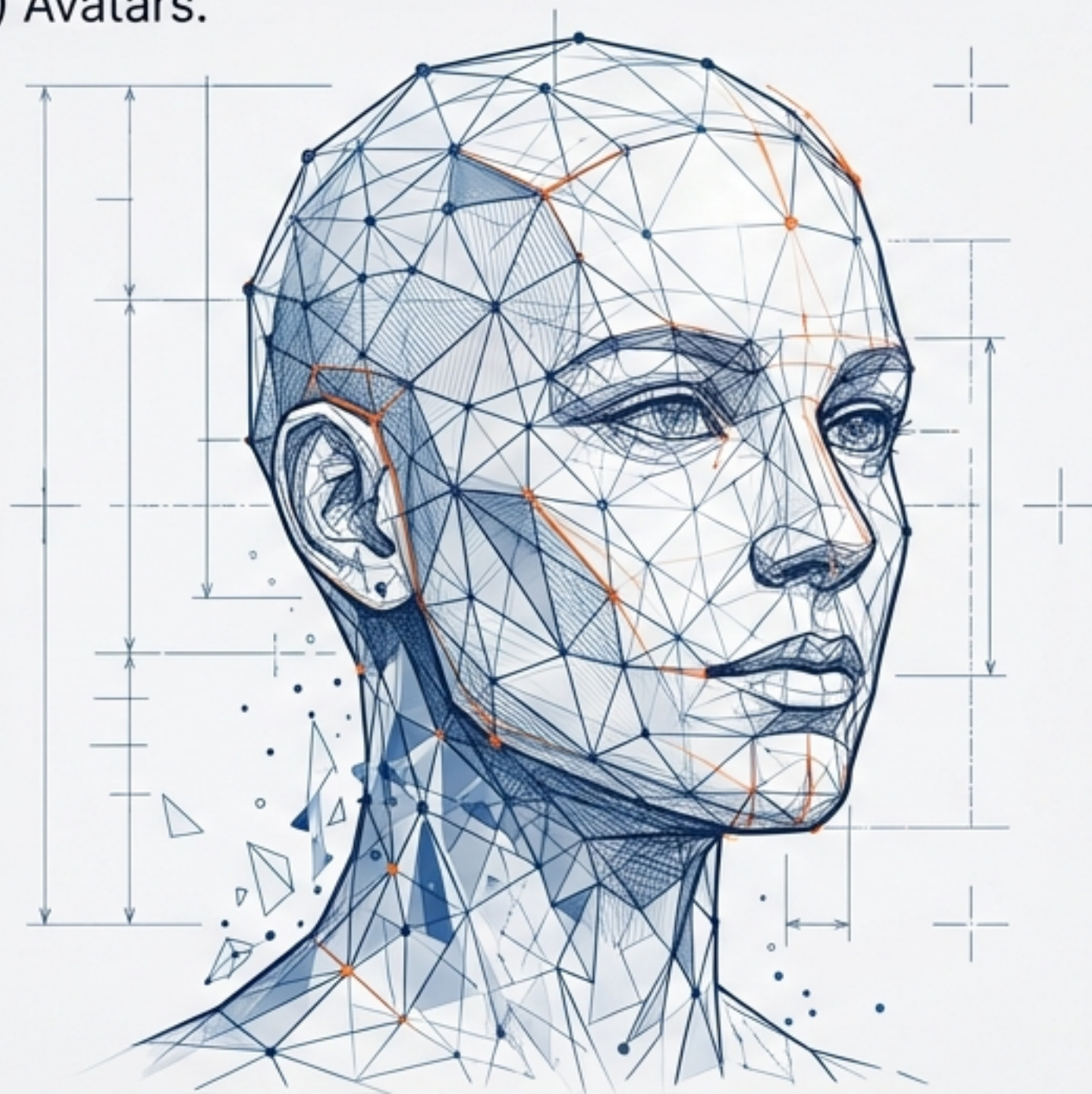


R3F Virtual Companion

Engineering an Innovation-Driven AI System

A rigorous Systems Engineering approach to integrating Large Language Models, Real-time Speech Synthesis, and React Three Fiber (R3F) Avatars.



SYSTEM SPECIFICATION

PROJECT:

Innovation-Driven AI

CONTEXT:

AI Systems Engineering

DATE:

February 2026

FRONTEND:

React Three Fiber / WebGL

BACKEND:

Node.js (Serverless)

INTELLIGENCE:

OpenAI LLM

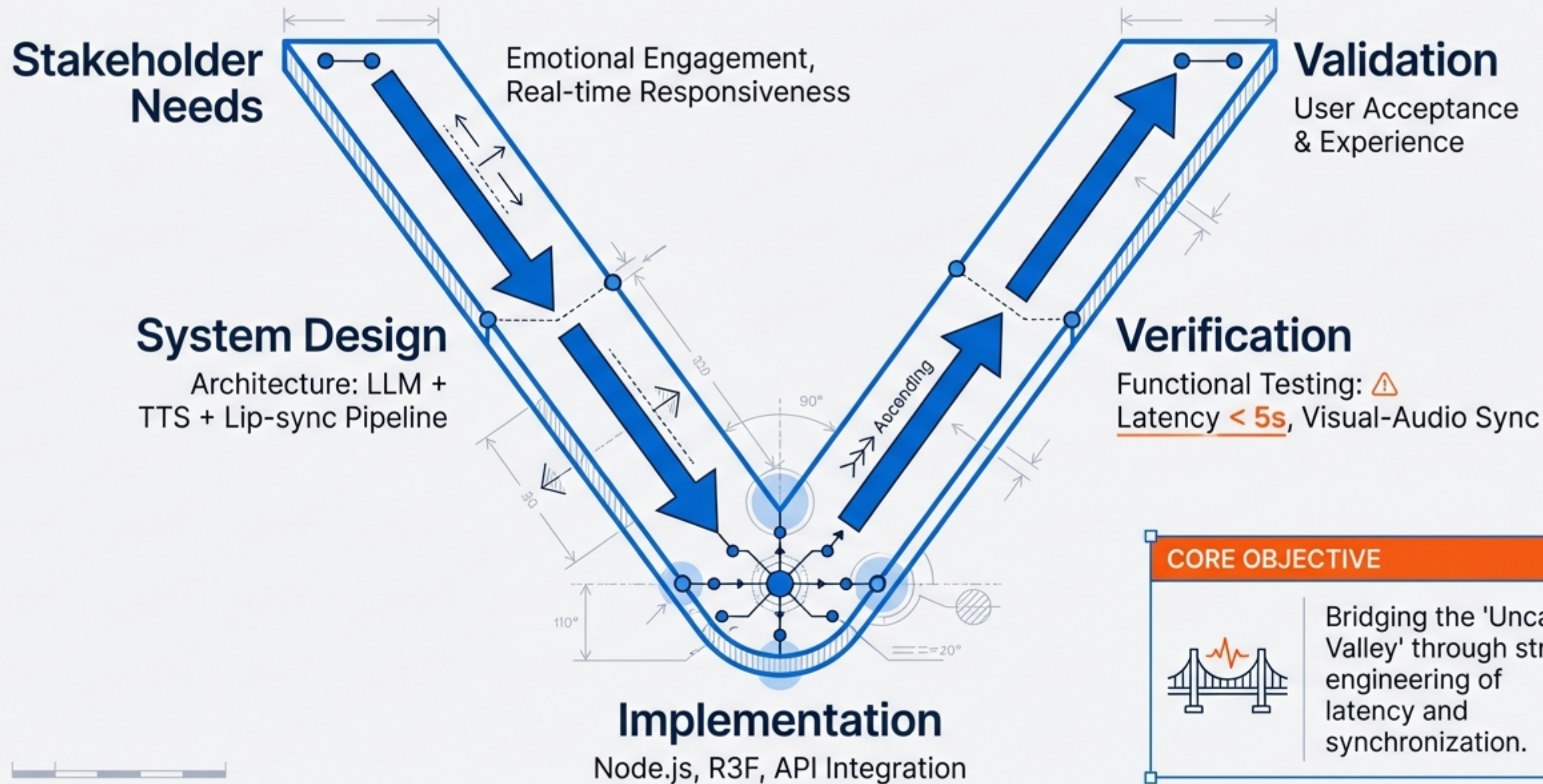
VOICE:

ElevenLabs TTS

SYNC:

Rhubarb Phoneme Analysis

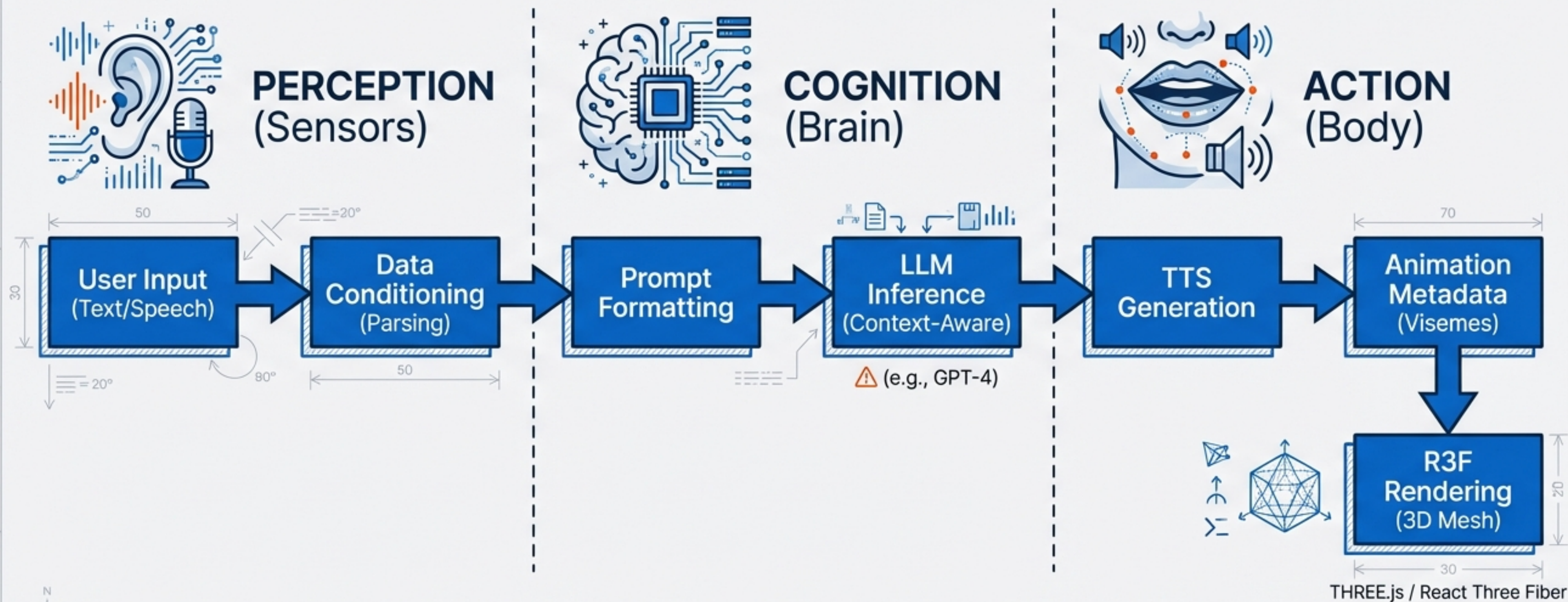
TITLE BLOCK	
Project code	Date



Functional Architecture: System Anatomy

An AI-Software-Centric Pipeline

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Project code	Date



Linear Data Flow: **Request** → **Processing** → **Render**

TITLE BLOCK	
Project code	Date

The diagram illustrates the AI Agent Architecture, divided into two main sections: the **INTELLIGENCE ENGINE** and the **OUTPUT CONSTRAINT (JSON SCHEMA)**.

INTELLIGENCE ENGINE:

- Model:** GPT-3.5 / GPT-4 via API
- Method:** Zero-shot / Few-shot Prompting
- Memory:** Sliding Context Window
- Transformer Model (LLM Layers):** Represented by a stack of five blue boxes with internal connections, receiving input from the Context Window and outputting a response.
- Context Window:** A blue box at the bottom, labeled "Context Window", with a width dimension of 50. It receives input from the left and feeds into the Transformer Model.

OUTPUT CONSTRAINT (JSON SCHEMA):

A code editor window displays a JSON schema for the output, with a width dimension of 70 and a height dimension of 50. The schema is:

```
1 {
2   "response": {
3     "text": "Hello! I am ready to help.",
4     "facialExpression": "happy",
5     "animation": "wave_hand",
6     "mood_intensity": 0.8
7   }
8 }
```

Strict JSON schemas prevent hallucinations and ensure frontend compatibility.

Data Management: The 4 Vs of Conversational AI

Inter

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VOLUME



Low Token Count per Request (<500).



High Throughput capability required.

VELOCITY



****CRITICAL CONSTRAINT****

End-to-End Latency < 2 seconds.

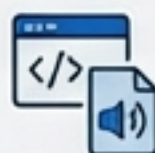


Real-time audio buffer.

VARIETY

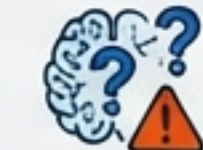


Input: Unstructured Natural Language.



Output: Structured JSON + Audio binary.
`{"text": "...", "audio": "..."}"`

VERACITY



Risk: Hallucinated emotion tags.



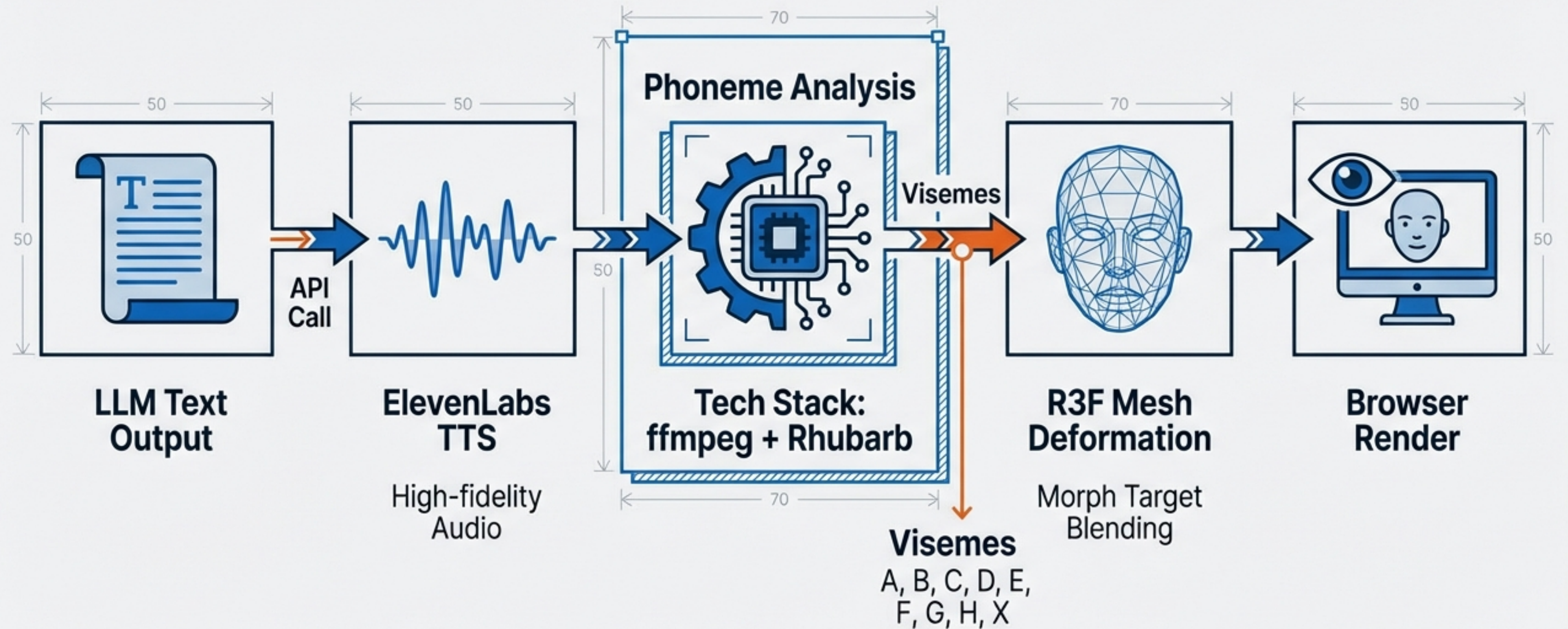
Mitigation: Strict Prompt Constraints.



FAIR Principles: Privacy-First Design. Data is accessible via API but not archived long-term.

Engineering the Body: Audio-to-Animation Pipeline

High-Fidelity Audio / Phoneme Analysis / Morph Target Blending



The Engineering Challenge: Latency & The Uncanny Valley

TITLE BLOCK

Project code

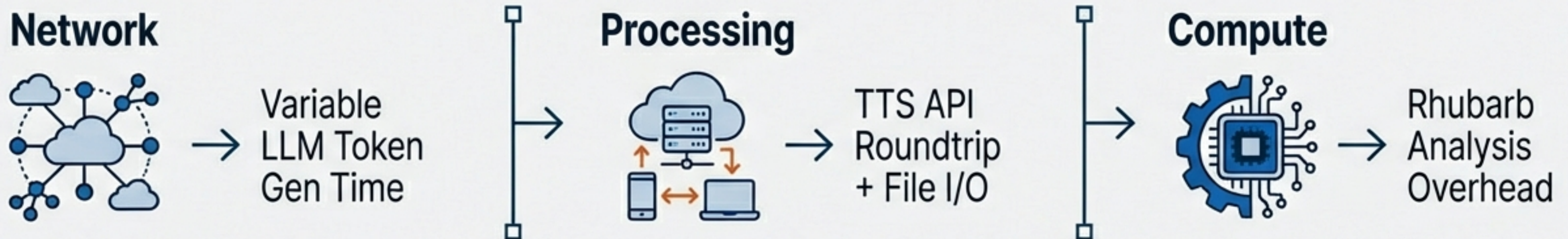
Date

The Conflict:



Desynchronization creates emotional disconnect.

Root Causes: System Bottlenecks



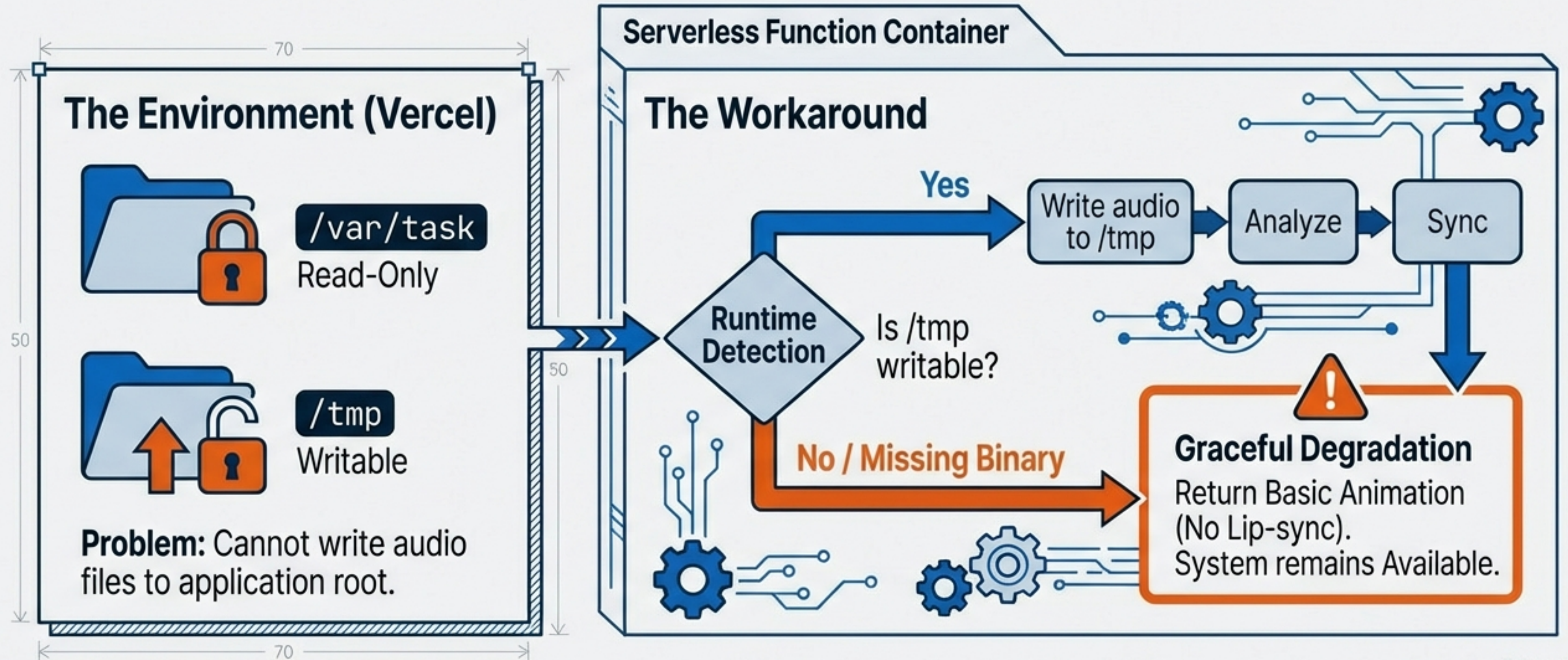
Overcoming Serverless Constraints

Filesystem limitations and Graceful Degradation

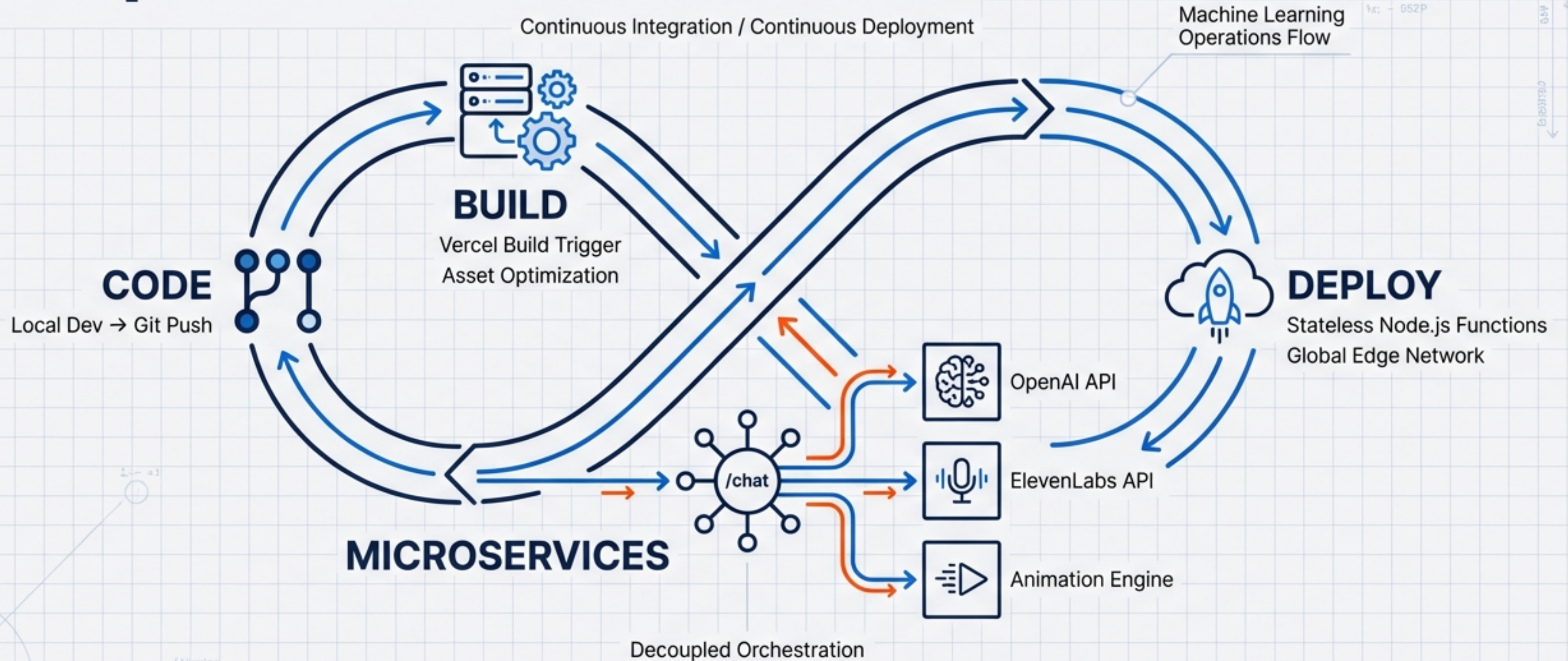
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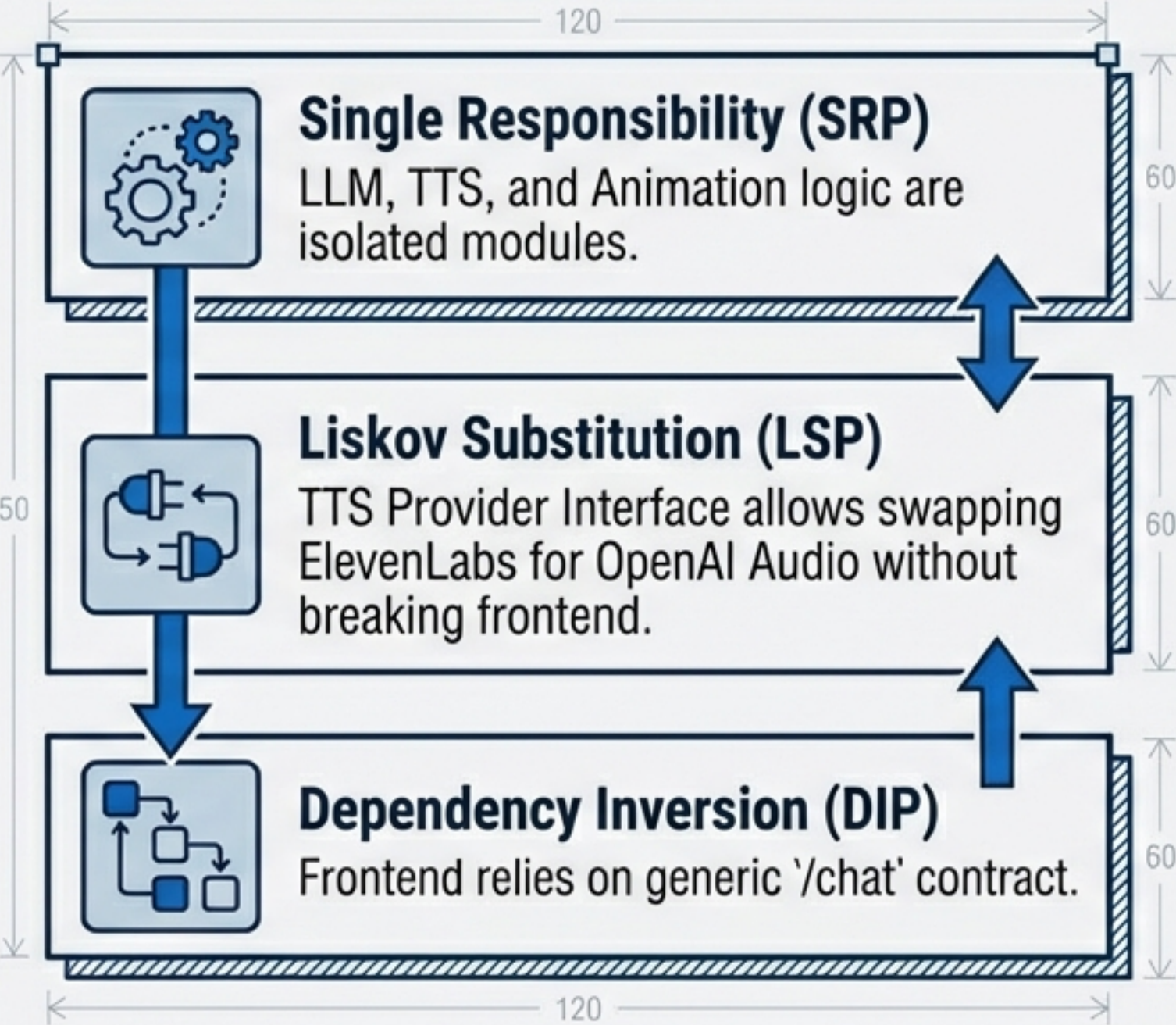
Operational Architecture: MLOps & CI/CD



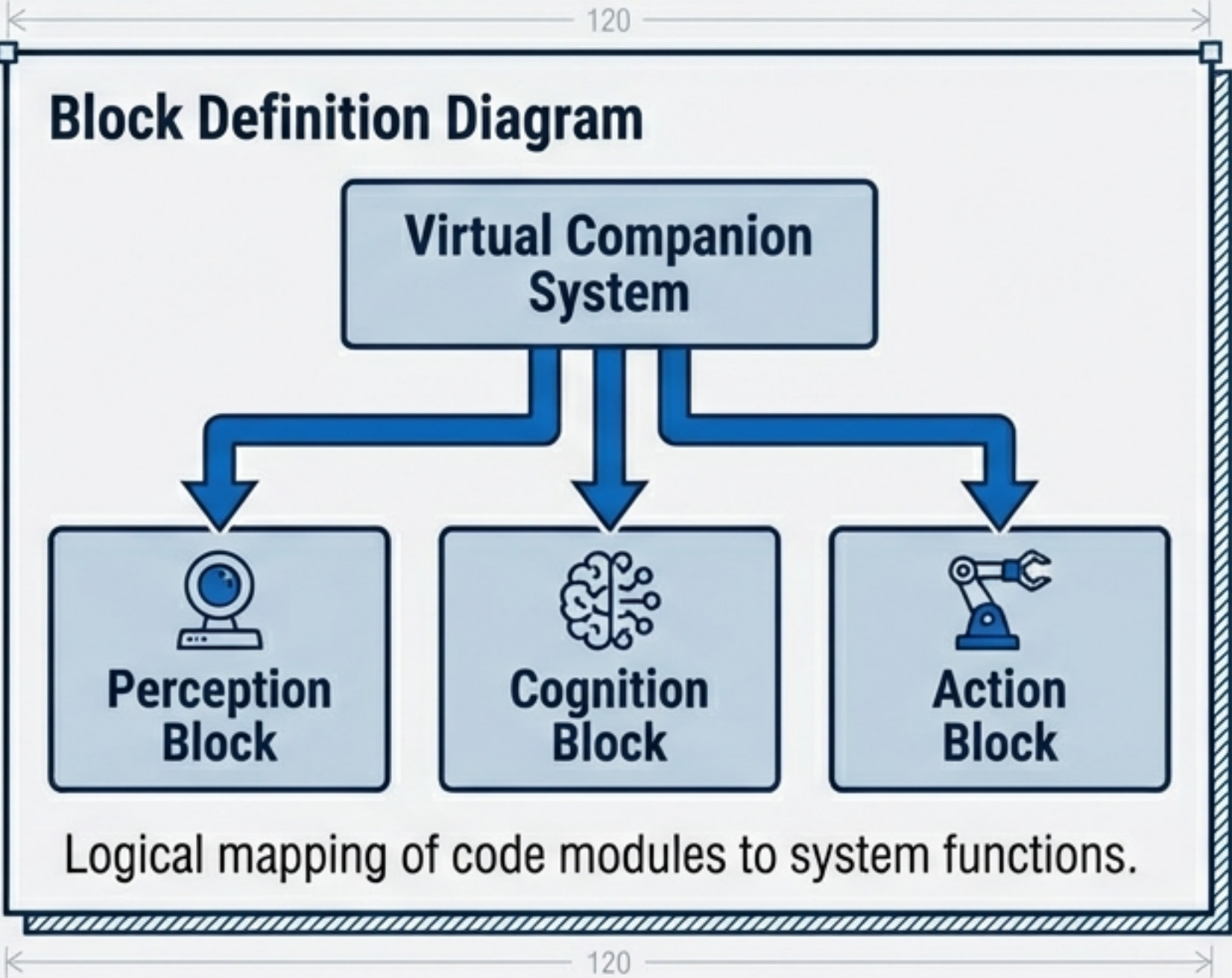
Engineering Foundations: SOLID & MBSE

TITLE BLOCK	
Engineering Foundations: SOLID & MBSE	
Project code	Date
Version 1.0	Oct 26, 2024

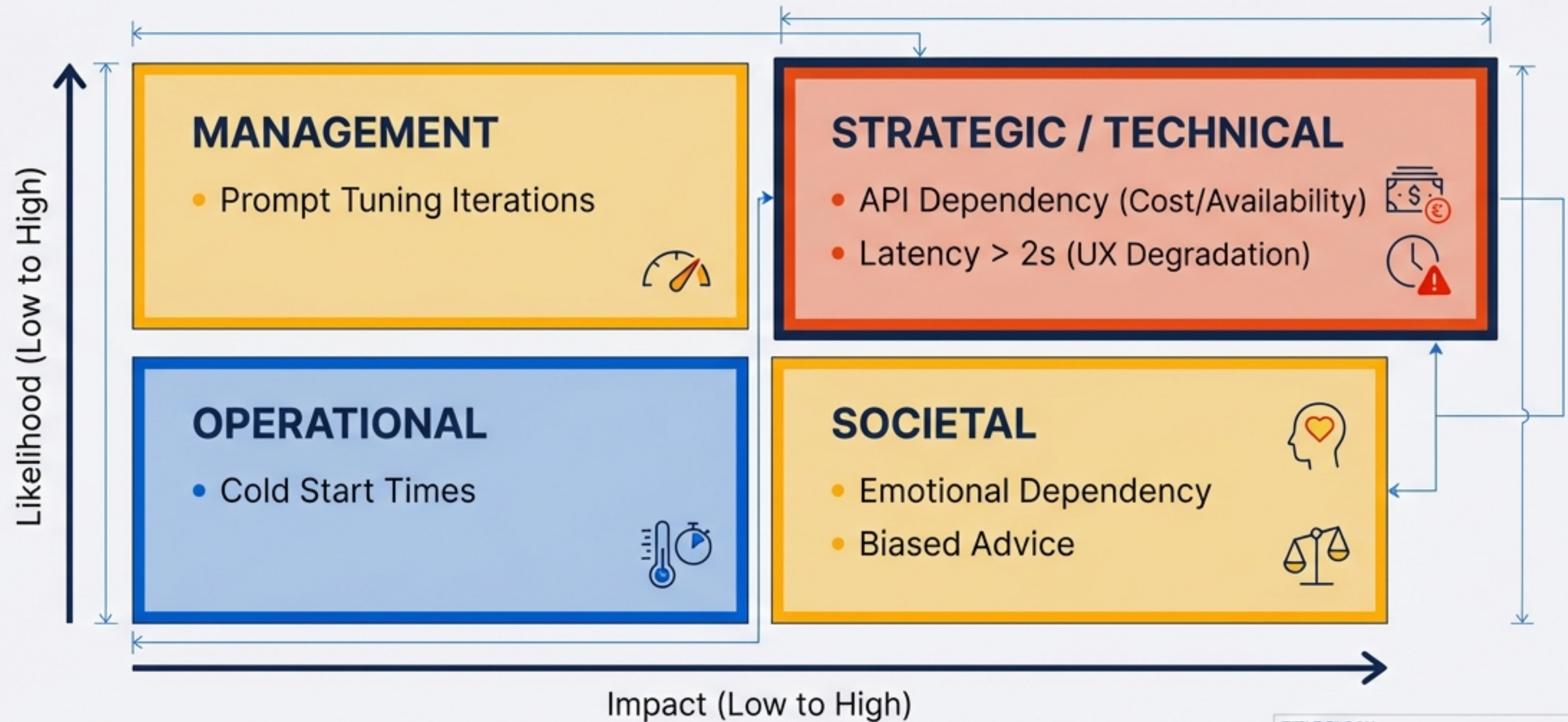
SOLID Principles



MBSE (Model-Based Systems Engineering)



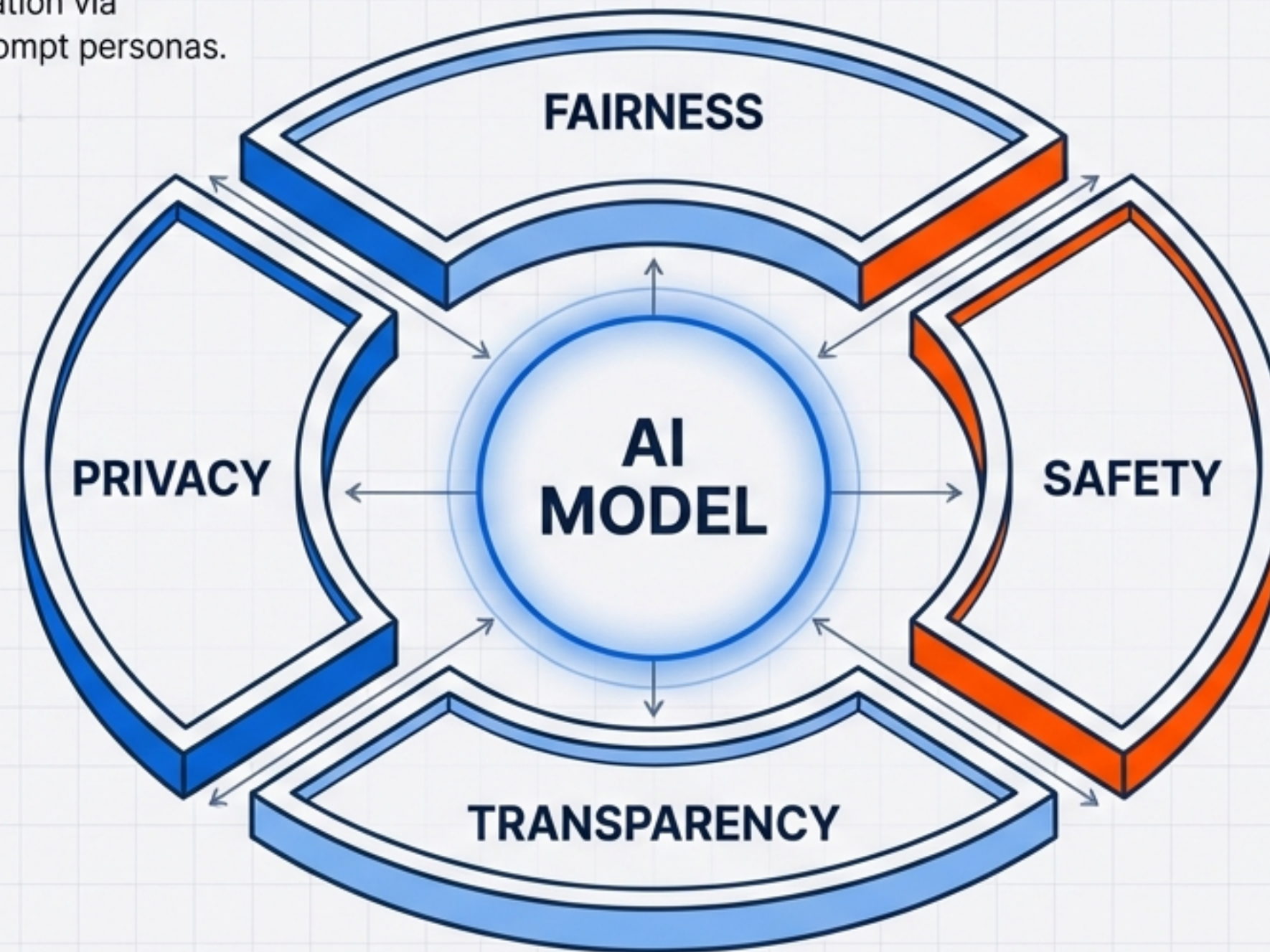
Risk Analysis Matrix



TITLE BLOCK		
Risk Analysis Matrix		
Project code: TBD	Version 1.0	Date: Oct 26, 2024

Trustworthiness & Ethics: FASTEPS Framework

- Bias Mitigation via diverse prompt personas.



- PRIVACY**
- No long-term storage.
 - Anonymized API calls.

- SAFETY**
- Guardrails against harmful/violent content.
 - Content Filtering.

Explicit metadata: "AI-Generated Response".
Visible Emotion Tags.

TITLE BLOCK: Trustworthiness & Ethics: FASTEPS Framework		
Project code: TBD	Version 1.0	Date:

Verification Strategy

Checklist Dashboard



Functional Testing

Valid JSON Structure (Text, Audio, Visemes).



Latency Evaluation

Target: < 5 Seconds. Status: PASS.



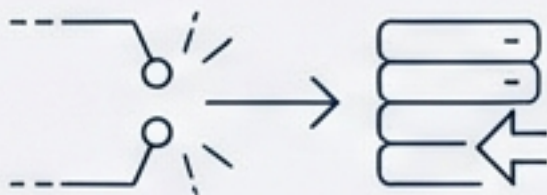
Closed-Loop Verification

Visual-Audio Alignment (Lip-sync accuracy).



Edge Case Handling

Missing API Keys / Read-Only FS -> Graceful Fallback.



Future State: Automated Regression Suites for Prompt Updates.

TITLE BLOCK:
Verification Strategy

Project code:
TBD

Version
1.0

Date:

Prototype Reality

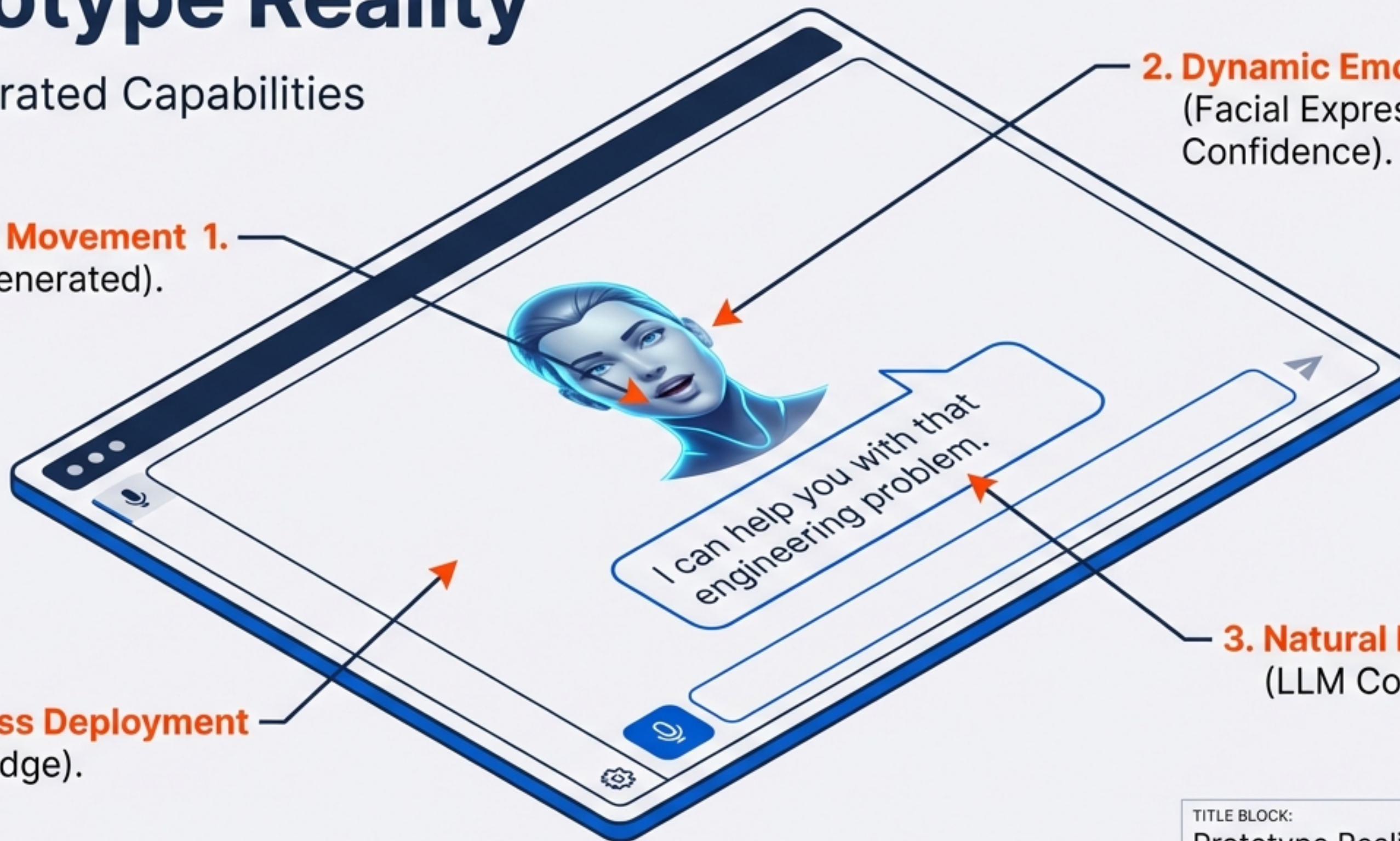
Demonstrated Capabilities

1. Synced Lip Movement
(Rhubarb generated).

2. Dynamic Emotion
(Facial Expression:
Confidence).

4. Serverless Deployment
(Vercel Edge).

3. Natural Dialog
(LLM Context Aware).



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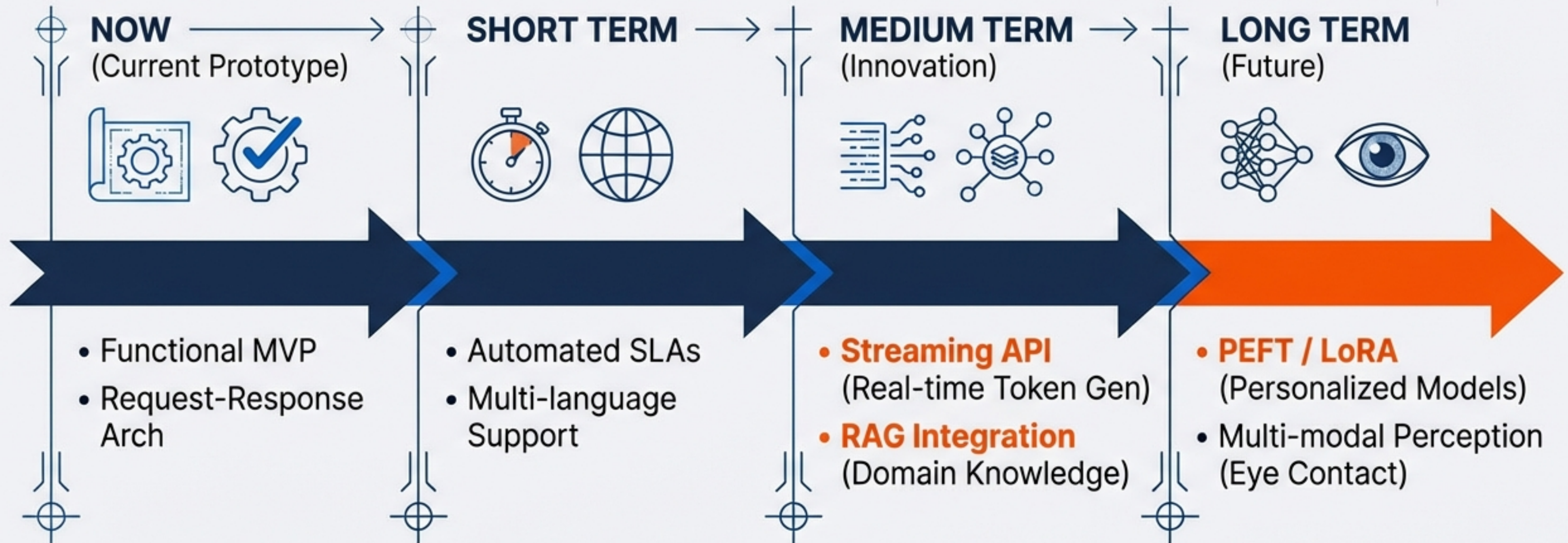
Prototype Reality

Project code:
TBD

Version:
1.0

Date:
Oct 26, 2024

Innovation Horizon & Roadmap



Conclusion: The R3F Virtual Companion represents a scalable, engineered foundation for immersive Human-Computer Interaction.